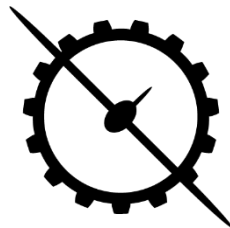


SKYLINE SENTRY



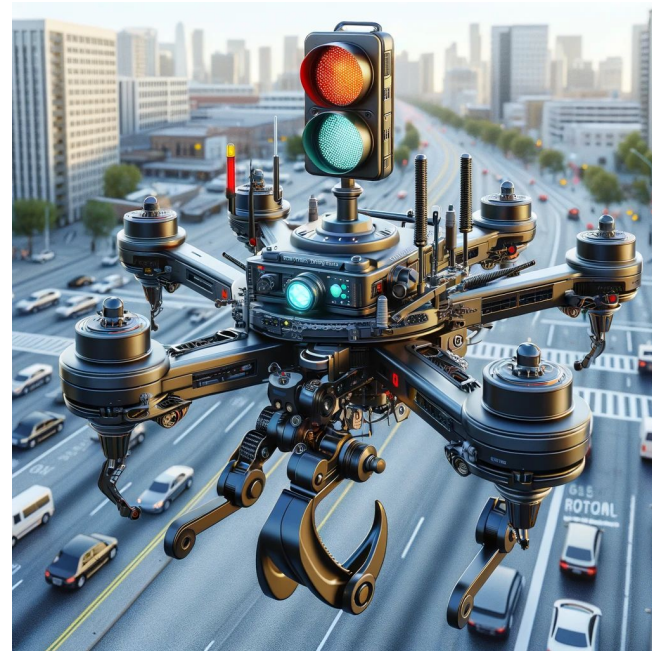
**AIAA Presentation -
Presenter Karthik Kannan**

April 4th/5th, 2024

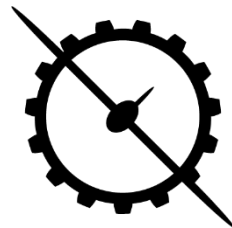


Introduction to the Skyline Sentry

The Skyline Sentry is a multi-functional autonomous power line retrieval drone with in-flight charging for enhanced emergency response and smart city applications. (The paper title) The drone does not exist in the capacity of the presentation but incorporates multiple ideas that could provide an innovative solution to a problem that exists today.



These image are AI generated



Problem Statement / The Oregon Tragedy

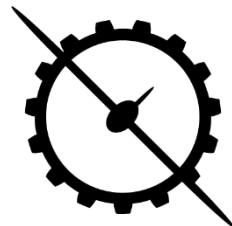
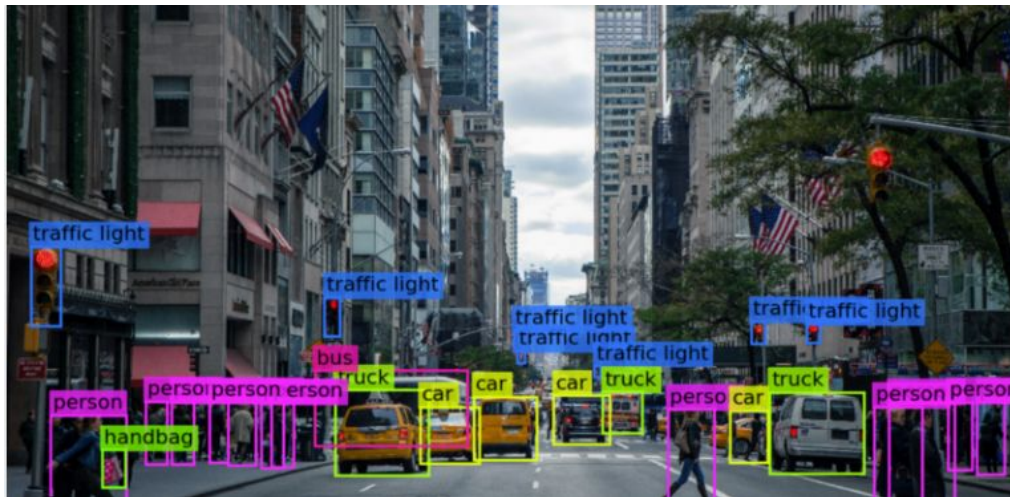
- Urban Safety Concerns
- Tragic Incidents
- Inadequate Traditional Methods
- Need for Innovation



Drone System Overview

The Core Integrations of the Sky Sentinel

- Autonomous Navigation
- In-Flight Charging Capability
- Advanced Computer Vision
- Emergency Situation Addressing
- Smart City Applications Enhancement



Autonomous Navigation

What does Autonomous Navigation mean?

There are 4 main components

- Environmental Awareness
- Path Planning
- Adaptive Learning
- Sensor Fusion



Obstacle Avoidance

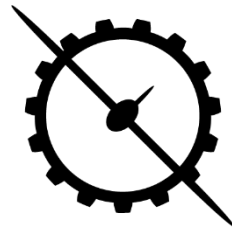
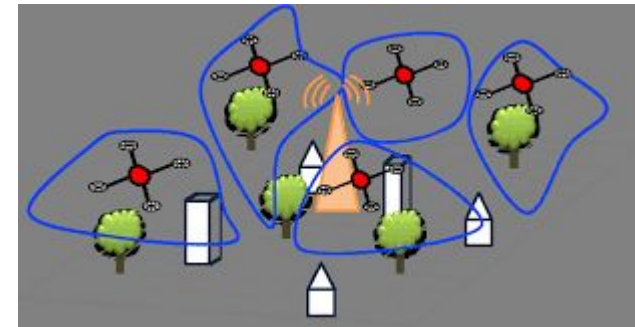
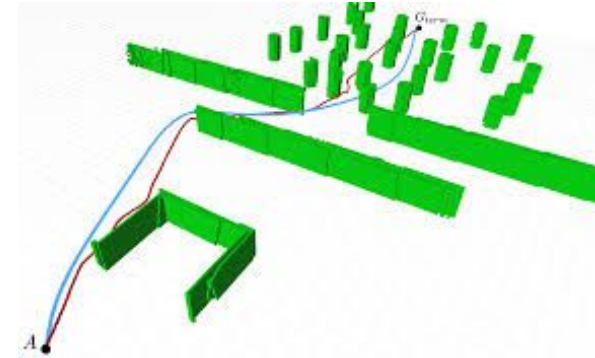
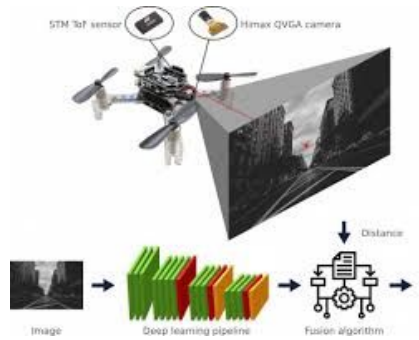
- Real-time processing for immediate obstacle detection and flight path adjustments.

Global and Local Path Planning

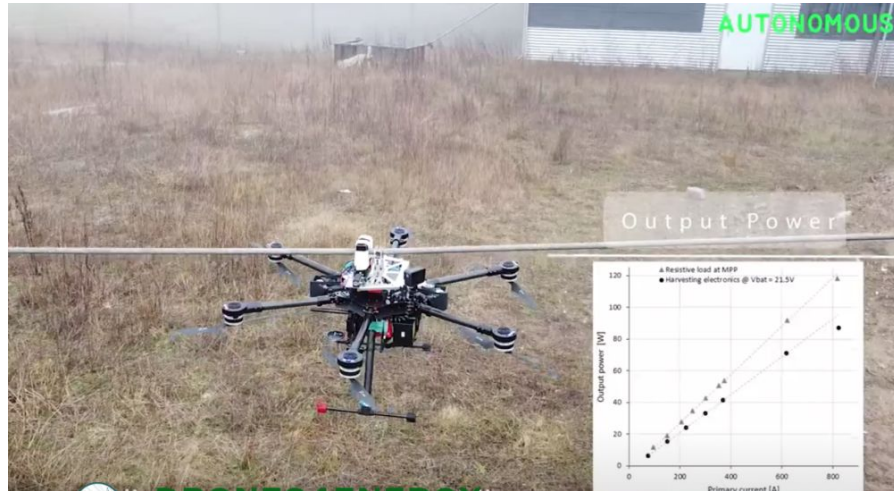
- Strategic and immediate routing decisions optimize flight efficiency and safety.

Autonomous Decision-Making

- Independent navigation decisions based on real-time data and pre-programmed logic.

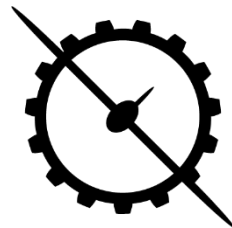
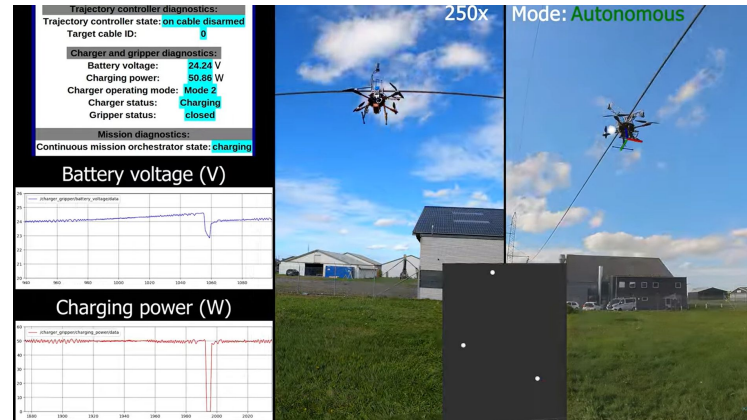


Power Line Charging Technology



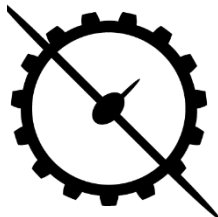
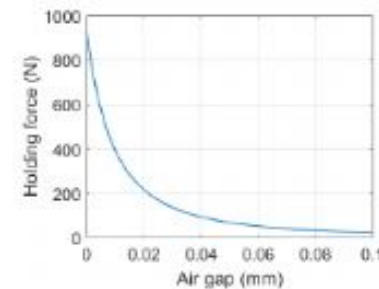
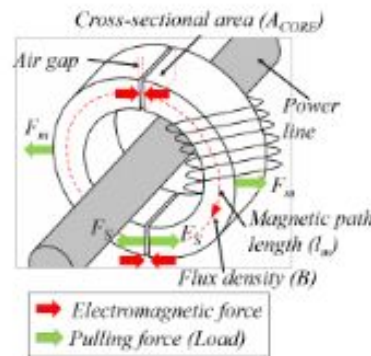
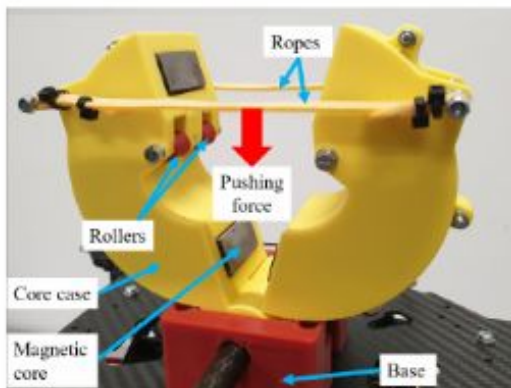
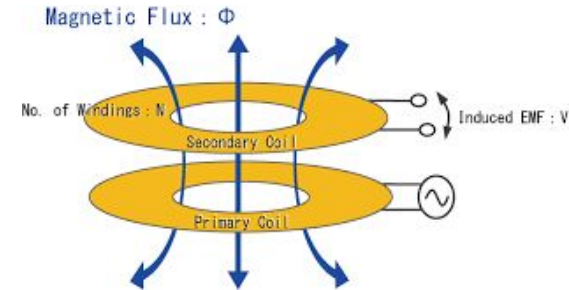
- **Groundbreaking Concept:** Introduces the ability for drones to autonomously dock with and recharge from live power lines, significantly extending operational capabilities.
- **Inspiration and Credit:** This innovative charging solution is inspired by pioneering research and developments in the field. Notable contributions by researchers like Boaz Ben-Moshe and the University of Southern Denmark.

- **Safety Mechanisms:** Advanced safety protocols and mechanisms to prevent electrical hazards during the docking and charging process
- **Autonomous Docking:** Utilizes algorithms and sensors to identify suitable power lines for charging



Innovative Grasping Clamps

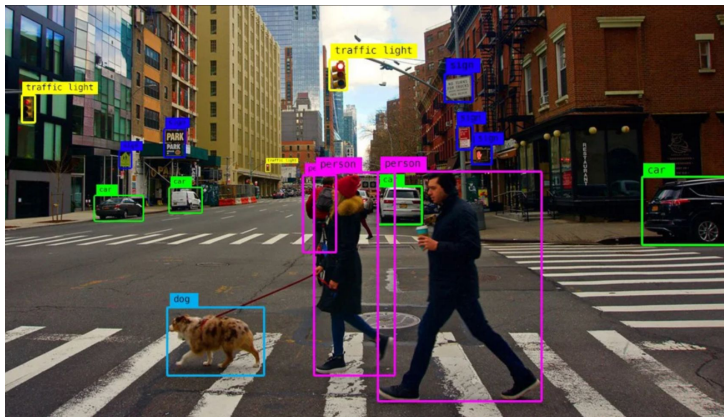
- **Advanced Design:** Engineered for precision and adaptability, enabling secure attachment to and interaction with various urban structures, including power lines.
- **Multi-functional Utility:** Beyond charging, clamps facilitate detailed inspections and minor repairs, enhancing the drone's utility in urban maintenance.
- **Durability in Urban Conditions:** Constructed from materials resistant to weather and electrical hazards, ensuring long-term operation in diverse urban environments.
- **Contribution to Emergency Response:** Enables drones to quickly reach and assess situations like downed power lines, improving the speed and safety of emergency responses.



Computer Vision

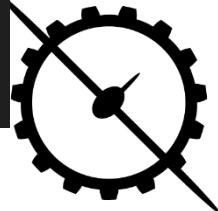


Unfortunately,
Computer's do have
better vision than us.
Might as well use it to
the best of our abilities.



```
def prepare_single_video(frames):
    frames = frames[None, ...]
    frame_mask = np.zeros(shape=(1, MAX_SEQ_LENGTH,), dtype="bool")
    frame_features = np.zeros(shape=(1, MAX_SEQ_LENGTH, NUM_FEATURES), dtype="float32")

    for i, batch in enumerate(frames):
        video_length = batch.shape[0]
        length = min(MAX_SEQ_LENGTH, video_length)
        for j in range(length):
            frame_features[i, j, :] = feature_extractor.predict(batch[None, j, :])
            frame_mask[i, :length] = 1
    return frame_features, frame_mask
```

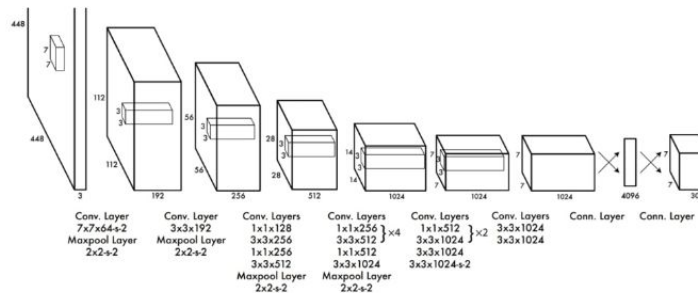
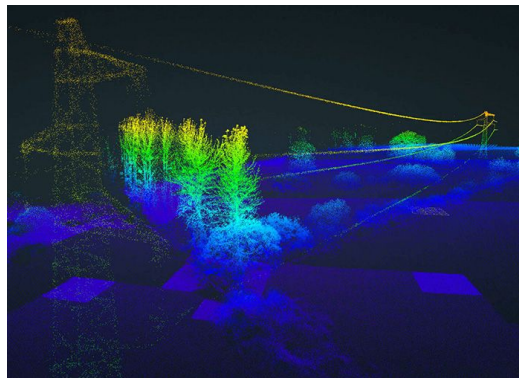
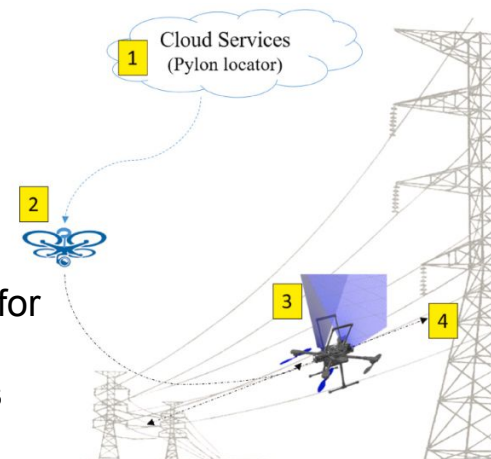






Yolov8 - The Vision

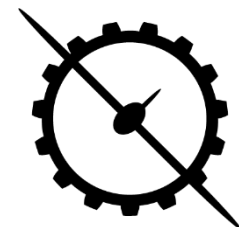
- **Latest in Object Detection:** YOLOv8 introduces unparalleled speed and accuracy, offering real-time identification in complex environments.
- **Deep Learning Innovation:** Utilizes an advanced convolutional neural network for superior accuracy in recognizing a wide array of urban objects and hazards.
- **Real-Time Hazard Detection:** Specially trained to identify critical urban hazards like downed power lines, enhancing emergency response and public safety.



The Architecture. Our detection network has 24 convolutional layers followed by 2 fully connected layers. Alternating 1×1 convolutional layers reduce the features space from preceding layers. We pretrain the convolutional layers on the ImageNet classification task at half the resolution (224×224 input image) and then double the resolution for detection.



- **Urban Adaptability:** Excelling in various lighting and settings, ensuring consistent performance across diverse urban landscapes.
- **Efficient Operations:** Facilitates focused deployment of resources by pinpointing areas of need, optimizing emergency and maintenance responses.
- **Autonomy in Solutions:** Empowers drones with autonomous decision-making, reducing human oversight and increasing operational efficiency.



Enhancing Operational Endurance

What are the main weaknesses of a modern drone?

- 1) Limited Battery life
- 2) Vulnerable to the environment
- 3) Human error

By using the aforementioned theory and implementation we are able to maximize our flight efficiencies and results.



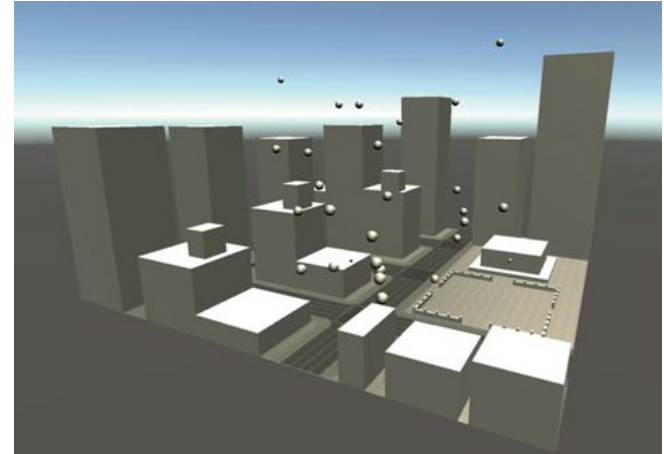
- 1) The power line charging idea helps us prolong lasting flight to not only complete its mission but also carry the power lines away from danger for as long as possible
- 2) With the use of computer vision it is able to detect obstructions and facilitate optimize pathing and choices that a human controlled drone may find difficult to asses
- 3) Human error is inevitable, but with a computer not only making decisions, it will also have better eyes and senses that will operate for maximum efficiency.

Drone Swarming

Our next main weakness comes in humans, we can't be everywhere at once, but build enough drones with a common objective and you can essentially be everywhere at once.

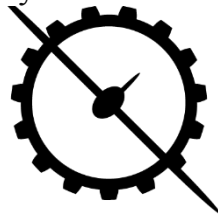
What is Drone Swarming?

Each drone in the swarm shares data and decisions in real-time, enhancing collective intelligence and adaptability to dynamic environments.



We maximize not only in Coverage but Speed as well. Because it is using computer vision, more information is better. It can also comprehend and make more rational decisions, such as identifying people in need of help, identifying weaknesses in infrastructure, fallen power lines, optimal pathing for humans as well.

It is easily scalable, it is not a one time emergency use it can be used in a multitude of facets.



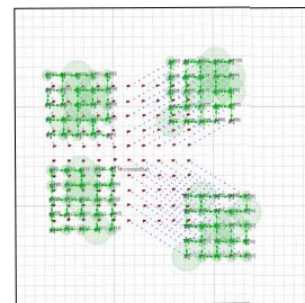
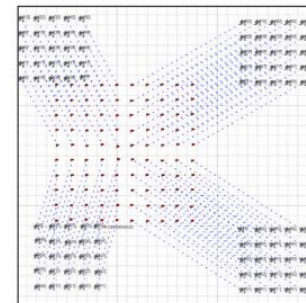
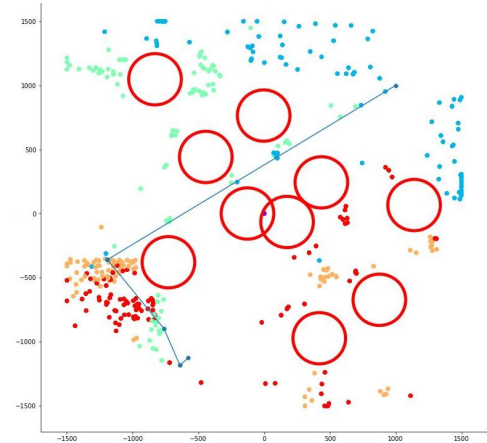
PSOP and Merging Algorithm

PSOP (Particle Swarm Optimisation Pathfinding) Algorithm:

- Designed to enhance drone swarming efficiency, especially in dynamic pathfinding.
- Improves autonomous UAV navigation in unknown or changing environments.
- Enables drones within a swarm to share information and optimize their paths effectively.

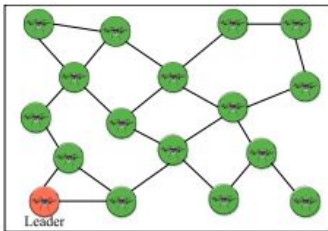
Drone Swarm Merging Algorithm:

- Allows drone swarms to operate either as individual units or in select groups.
- Facilitates collaborative effort among drones, covering large areas quickly.
- Extremely effective for emergency decision-making and disaster response scenarios.

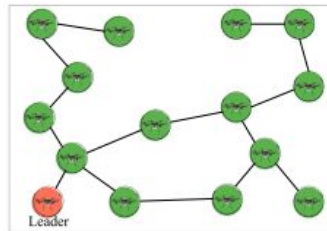


(a) Initial configuration of four drone swarms

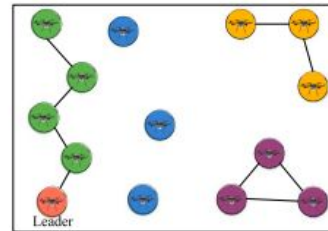
(b) The process of merging four drone swarms



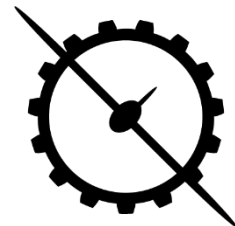
(a) The topology of the drone swarm



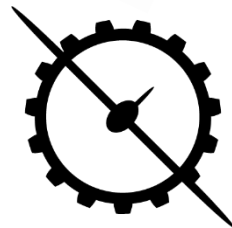
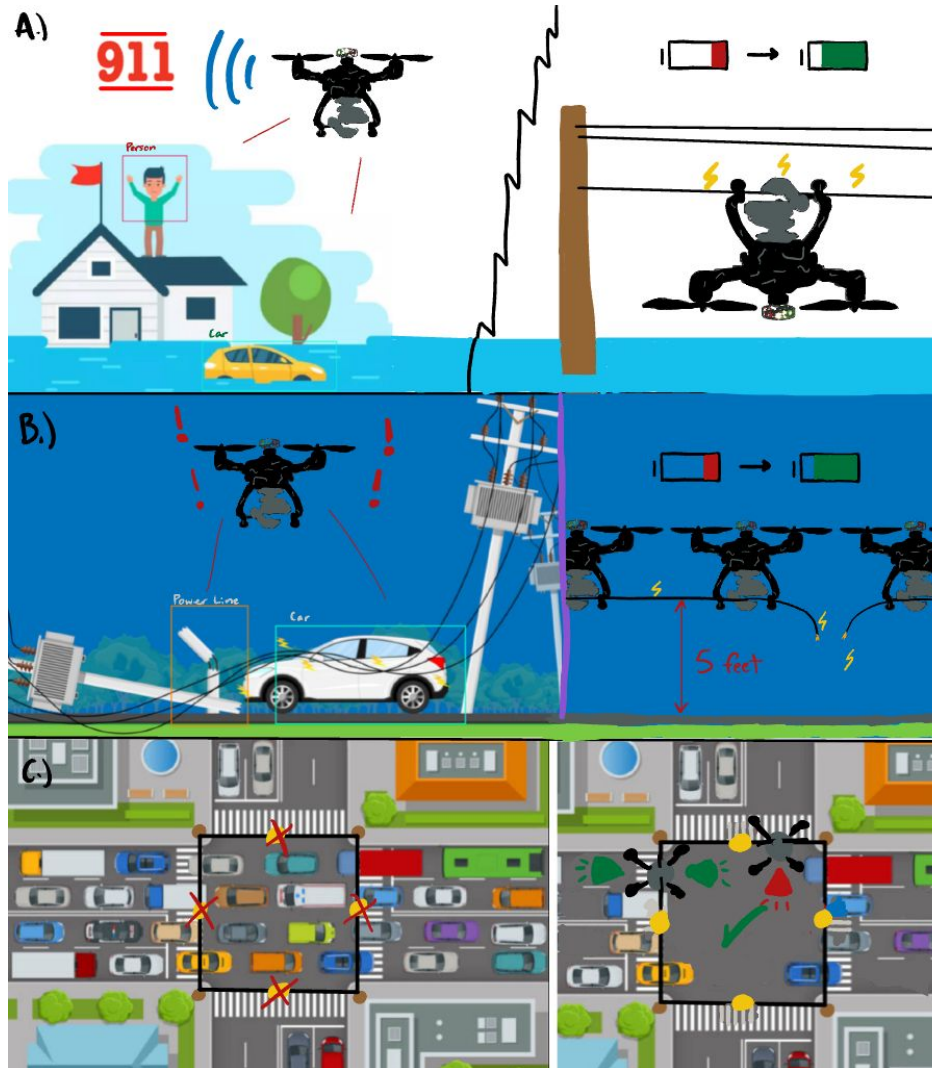
(b) The topology changing with drones lost



(c) The topology splitting with drones lost



Concept of Operation (Hand Drawn)



Search and Rescue Operations

Situation Overview:

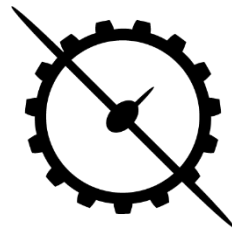
- Following a natural disaster, a large urban area requires immediate search and rescue operations to locate survivors and assess the extent of damage.

Drone Swarm Activation:

- Drones equipped with thermal imaging and computer vision technologies are deployed to scan the disaster-stricken area, guided by the PSOP algorithm for efficient search patterns.

Operational Execution:

- The swarm identifies heat signatures and potential survivors, marking their locations for rescue teams.
- Obstacles and safe paths are mapped out in real-time, assisting ground rescue operations in navigating the affected area.
- Continuous aerial surveillance provides ongoing assessment of the situation, enabling dynamic adjustment of rescue efforts and resource allocation.



Managing Fallen Power Lines

Situation Overview:

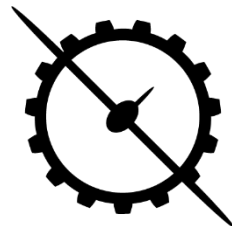
- A severe disaster causes multiple power lines to fall in a densely populated urban area, posing significant risks to residents and disrupting critical infrastructure.

Drone Swarm Activation:

- A swarm of drones equipped with the PSOP algorithm and computer vision technology is immediately deployed to the affected area.

Operational Execution:

- Drones autonomously navigate through the urban environment, identifying downed power lines and hazardous areas with high precision.
- Real-time data collected by drones is relayed to emergency services for efficient, informed response planning.
- Temporary interventions are initiated by drones with grasping clamps to mitigate immediate dangers, ensuring public safety until repair crews arrive.



Smart City Application

Situation Overview:

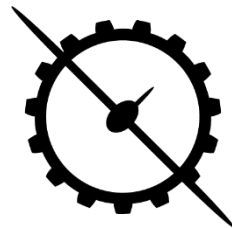
- Regular inspection of urban infrastructure, including bridges, buildings, and power lines, is required to maintain city operations and prevent incidents.

Drone Swarm Activation:

- A drone swarm is deployed over a designated urban area, utilizing autonomous navigation and swarming techniques to cover extensive zones efficiently. Using power lines essentially as it roads and paths.

Operational Execution:

- Drones perform detailed inspections using advanced imaging and sensing technologies, identifying potential issues or degradation in infrastructure.
- Collected data is analyzed using computer vision algorithms, providing actionable insights for maintenance priorities and planning.
- Swarming capabilities allow for rapid, comprehensive coverage, significantly reducing inspection times and costs.



Safety and Regulatory Concerns

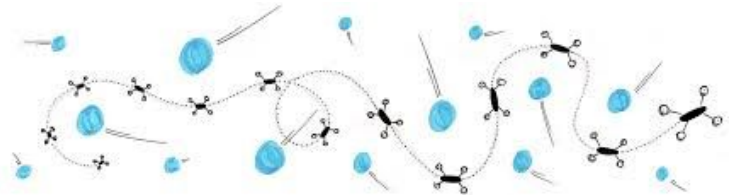
Safety Risks

- Collision Hazards:

- Problem: Potential collisions with buildings, temporary structures, or other drones.
- Drone Solution: Utilizes advanced computer vision and the PSOP algorithm for dynamic, real-time obstacle avoidance, reacting faster than human operators.

- Falling Debris:

- Problem: Risk of drones dislodging unstable debris from damaged structures.
- Drone Solution: Employs sensor technology to assess structural stability before close inspection, minimizing interaction with potentially unsafe debris.



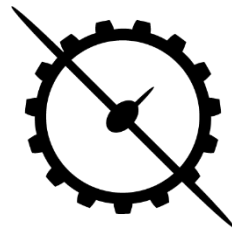
Privacy and Data Security

- Privacy Invasions:

- Problem: Inadvertent capture of private moments due to cameras on drones.
- Drone Solution: Implements smart image processing algorithms to blur or anonymize individuals automatically, protecting privacy.

- Data Security:

- Problem: Risk of unauthorized data access or misuse.
- Drone Solution: Uses encrypted communication and data storage protocols to secure collected information from interception or unauthorized access.



Environmental Impact and Ethical Dilemma

Environmental Impact



Noise Pollution:

- Problem: Increased drone usage contributing to noise pollution.
- Drone Solution: Designs drones with noise-reduction technologies, minimizing sound emissions during flight.

Impact on Wildlife:

- Problem: Potential disturbance to urban wildlife, particularly birds.
- Drone Solution: Utilizes wildlife detection algorithms to avoid areas with high wildlife activity, reducing disturbance.



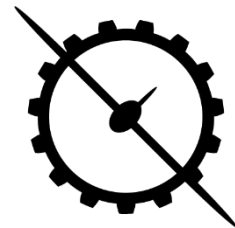
Ethical Considerations

Prioritization of Areas:

- Problem: Ethical dilemmas in deciding which areas to survey or rescue first.
- Drone Solution: Implements objective criteria based on urgency and impact, supported by AI analysis, to make unbiased prioritization decisions.

Dependency and Response Delay:

- Problem: Over-reliance on drones potentially delaying immediate human response.
- Drone Solution: Designed to augment rather than replace human responders, providing critical support and information to enhance overall response efforts.



Future Developments

Advanced Autonomy and Intelligence:

Develop AI for fully autonomous missions and cognitive computing for dynamic reaction to urban changes.

Enhanced Communication Systems:

Implement 5G technologies for improved drone coordination and secure channels for data integrity.

Improved Energy Solutions:

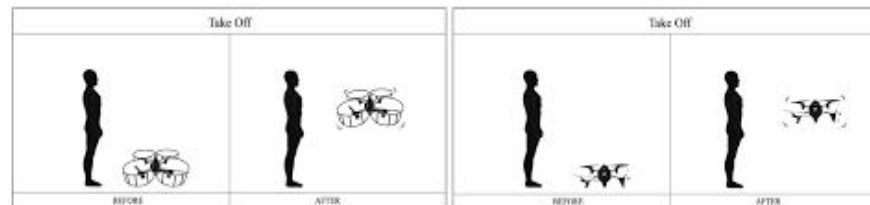
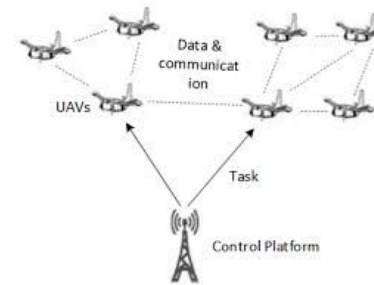
Explore sustainable energy sources like solar power and energy harvesting to extend flight times.

Robust Regulatory Frameworks:

Collaborate on global drone operation standards and ethical guidelines for privacy and data protection.

Human-Drone Interaction:

Create user-friendly interfaces for wider accessibility and establish public interaction protocols.



Potential Partners and Stakeholders

Government and Regulatory Bodies:

- Local municipalities for urban integration, and national aviation authorities like the FAA for regulatory compliance.

Technology and Industry Partners:

- Leading drone manufacturers for cutting-edge hardware, and telecommunications companies for essential connectivity solutions.

Research and Academic Institutions:

- Universities with advanced robotics labs offering innovation and research collaboration opportunities.

Private Sector and Industry:

- Utility companies in need of efficient infrastructure inspection and maintenance services.

Public and Community Organizations:

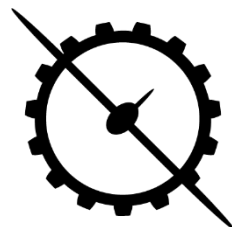
- Environmental NGOs leveraging drones for conservation efforts and environmental monitoring.



OPTELOS



CHCNAV





SKYLINE SENTRY



Innovative Urban Safety Solution: Skyline Sentry introduces cutting-edge drone technology to tackle urban safety challenges, focusing on fallen power line management and emergency response.

Advanced Technological Integration: Combines autonomous navigation, in-flight charging, and advanced computer vision with YOLOv8, enhancing operational efficiency in urban environments.

Swarming for Comprehensive Coverage: Utilizes drone swarming and PSOP algorithm for extensive, adaptable urban coverage, setting a new standard in collaborative drone operations.

Safety and Environmental Commitment: Emphasizes rigorous safety protocols, privacy protection, and minimal environmental impact, demonstrating responsible technological advancement.

Vision for Smarter Cities: Embodies the potential of engineering and technology to create safer, smarter urban landscapes, highlighting a commitment to societal progress and innovation.



Questions?

